

To understand the risk exposure of these strategies, we run factor regressions on the trend-following reversal returns against a series of standard financial market risk factors. The regression output reveals that the trend-following reversal returns are closely related to the market as well as the momentum factors. However, certain strategies, especially the contrarian loser, produce significant alphas which are not explained by the popular risk factors. Moreover, the contrarian loser represents an effective strategy for investors to avoid momentum risks, owing to its low correlation with trend-following factors.

In short, the contribution of this paper to the literature is threefold. First, we find that time series reversal usually occurs between 12 and 24 months after the formation of trend-following strategies, which is much shorter than the findings documented in the cross-sectional reversal literature. Second, we determine which of the trend-following components contributes most to the reversal profits; this finding contradicts the prominent behavioural theory of [Hong and Stein \(1999\)](#). Third, we document a trend-following reversal strategy earning significant abnormal returns based on both time series continuation and reversal.

The rest of the paper is organised as follows. In Section 2 we describe our data collection and transformation methods, and explain the intuitions of different trend-following trading signals. Section 3 presents the performance results of multiple trend-following strategies and uncovers the timing of time series reversal. In Section 4, we perform two analyses—sign analysis and sub-portfolio performance evaluation—to investigate the reversal property of each subgroup via a decomposition of trend-following strategies. Section 5 introduces the trend-following reversal strategy and explores its factor loadings. Finally, Section 6 summarises the conclusions.

2. Data and Trading Signals

2.1. Future Contracts and Other Data

We collected data for 55 of the world's most liquid exchange traded futures from January 1985 to March 2015. Such a dataset is similar to that used in TSM studies